MC 451 Codebook and Qualitative Memo: Worked Example

- Rebuilt from my own research project, Play to Learn in VR (P01.01), continuing the student study from the Project Prospectus and Definitions Practice examples: one coder reading 150 tweets from 2021 about playing in VR.
- Everything under "The example" is what a student would submit: the codebook and the memo as one document with two headed parts. Every example tweet is paraphrased from a real tweet in the archive; none is quoted, so a search cannot lead back to the account.
- The rubric map at the end shows which grading row each part earns and why.

The example

Part 1: Codebook

Study overview

This codebook measures whether tweets about playing in virtual reality (VR) that refer to learning are more often positive toward VR than tweets that do not, and which benefits and obstacles of VR those learning tweets name. The material is English-language tweets, not retweets, that mention VR together with play, drawn from the Play to Learn in VR archive collected through the Twitter Academic Research API. The time period is calendar year 2021, January 1 through December 31.

Unit of analysis

One tweet: the text the author wrote in that single post. A quoted tweet, a linked page, an image, and the replies underneath are not part of the unit. Every variable below is coded on one tweet. Each tweet is also recorded with three fields that are looked up, not coded: its tweet ID, its posting date, and its like count.

Variable 1: Learning reference

Element	Learning reference
Conceptual definition	Whether the tweet is about someone gaining or passing on a skill or knowledge in or through VR: learning, teaching, studying, practicing, or being trained.
Operational definition	Read the whole tweet. Mark 1 if any clause describes a person (the writer, a named person, learners in general, or a customer in an advertisement) learning, teaching, practicing, studying, or being trained in something, and the activity happens in or through VR. Otherwise mark 0. If the tweet contains a learning word, check what the word means in its sentence before marking.

Element	Learning reference
Level of measurement	Nominal: two named categories with no order between them.
Coded values	0 = no learning reference. 1 = learning reference.
Edge-case rules	1. A learning word inside a game, product, or film title, or inside an idiom (learned my lesson, hype train), is coded 0 unless another clause describes learning. 2. "Learn more" in a promotion, meaning "find out about," is coded 0. 3. An advertisement for a product that teaches or trains (a crane-operator simulator, a public-speaking practice app) is coded 1. The variable records that learning is described, not who describes it or why. 4. Machine-learning and artificial-intelligence hashtags are coded 0: the learning is a computer's, not a person's. 5. Learning to play a VR game, or relearning it on new controllers, is coded 1, because the skill is gained in VR.

Variable 2: Tone toward VR

Element	Tone toward VR
Conceptual definition	The writer's evaluation of VR or of the VR experience the tweet describes: whether the tweet speaks well of it, badly of it, or neither.
Operational definition	Find every word, phrase, or emoji in the tweet that evaluates VR, a VR headset, or a VR experience, including a game played in VR (fun, love, amazing; nauseous, broken, waste of money). Mark 3 when every evaluation is favorable, 1 when every evaluation is unfavorable, and 2 when the tweet has both or has none.
Level of measurement	Ordinal: three categories in order from negative to positive, with no claim that the steps between them are equal.
Coded values	1 = negative. 2 = neutral or mixed. 3 = positive.
Edge-case rules	1. Praise for a game played in VR counts as an evaluation of a VR experience. Praise for a game played outside VR, a streamer, or a friend does not count. 2. An advertisement that lists features or a release date with no evaluative word is coded 2. An advertisement that calls the experience amazing is coded 3, because the variable records what the tweet says, not whether the writer is sincere. 3. A complaint the writer laughs off is coded by its literal words, so it is 1 unless a favorable clause also appears, which makes it 2. 4. Emoji count as evaluations: a heart or a fire emoji attached to VR is favorable, and a vomiting face is unfavorable.

Variable 3: Affordance named

Element	Affordance named
Conceptual definition	The benefit the tweet attributes to VR: something VR lets a person do, or do better, in the writer's account.
Operational definition	Look for a clause that says what VR, a headset, or a VR experience lets someone do or makes better. Match that clause to one category below. If the tweet names benefits in two categories, mark the one that takes up more of the tweet's words; if they are equal, mark the one that appears first. If no clause names a benefit, mark 0.
Level of measurement	Nominal: named categories with no order.
Coded values	0 = no benefit named. 1 = fun or engagement (enjoyment, excitement, being absorbed). 2 = hands-on interaction (moving, building, handling objects, being inside the scene). 3 = doing it with others (playing, meeting, or learning alongside other people). 4 = practicing a skill (getting better at something, in a game or outside it, in a safe setting). 5 = other benefit (exercise, wellbeing, anything else).
Edge-case rules	1. A feature named without saying what it lets a person do (multiplayer, hand tracking, wireless play) is coded 0. A feature is not a benefit until the tweet says what it is for. 2. "Fun" or "love it" with no more detail is coded 1. 3. Learning to play a VR game better is coded 4, the same as practicing a clinical skill in VR, because the variable records the kind of benefit, not whether it carries outside the game. 4. A tweet that praises VR for company is coded 3 only when other people are involved; a solo experience described as absorbing is coded 1.

Variable 4: Constraint named

Element	Constraint named
Conceptual definition	The obstacle the tweet attributes to VR: something that stopped, limited, or discouraged the writer or someone else from doing something in VR.
Operational definition	Look for a clause that names a problem with VR, the hardware, or the experience that limited or discouraged use. Match it to one category below. If the tweet names problems in two categories, mark the one that takes up more of the tweet's words; if they are equal, mark the one that appears first. If no clause names a problem, mark 0.
Level of measurement	Nominal: named categories with no order.

Element	Constraint named
Coded values	0 = no obstacle named. 1 = physical discomfort (motion sickness, nausea, eye strain, headache, fatigue). 2 = hardware cost (the price of a headset or computer, cannot afford one). 3 = technical performance (lag, crashes, glitches, setup trouble, incompatibility). 4 = other obstacle (space, hygiene, other players' behavior, anything else).
Edge-case rules	1. A problem with something other than VR (a slow phone, a lost username) is coded 0 unless the tweet ties the problem to VR. 2. A warning or advice about a problem (how to build up tolerance to motion sickness) is coded in that category even when the writer did not suffer it, because the tweet names it as an obstacle. 3. A problem the writer jokes about expecting someday, without saying it has limited anyone yet, is coded 0. 4. "I can't play, no headset yet" is coded 2 only when price or affordability is stated; otherwise it is coded 4.

Decision rules for the whole codebook

These rules settle the cases that did not sort themselves while I read the first 30 tweets. Each one applies before or across the four variables.

1. Before coding, check that the tweet is about virtual reality. If VR appears only inside another word, a username, or an unrelated abbreviation, and the tweet is about something else (a sports position, a rhythm game played on a monitor), mark it out of frame, code nothing, and replace it with the next tweet on the reserve list.

2. If two tweets in the sample carry the same text word for word (a bot reposting one promotion), code the first one and mark the second a duplicate; it is replaced from the reserve list.

3. The archive strips digits and some punctuation, so a console name or a date can appear with its number missing. Code the words that remain and never fill in a missing word from a guess.

4. A tweet that is only a string of hashtags, with no sentence, is coded 0 on learning reference, 2 on tone, 0 on affordance, and 0 on constraint.

5. Code what the tweet says, not who sent it. An advertisement, a non-fungible token (NFT) promotion, or an adult-game listing that slipped past the archive's sensitive flag is coded by its words like any other tweet. A tweet whose text is sexually explicit is marked out of frame and replaced, and it is never used as an example.

6. When one clause fits two variables, code it on each. A sentence saying friendly strangers taught the writer to play is a learning reference (1) and an affordance of doing it with others (3).

7. Code the username, the follower count, and the like count hidden. The like count is looked up only after all four variables are coded for that tweet.

Examples

Two or three examples per category, each paraphrased from a tweet in the archive.

Category	Examples (paraphrased)
Learning reference 1	(a) A new headset owner says friendly strangers in a VR tag game taught them how to play.
	(b) A parent says a multiplayer VR puzzle game lets them and their son play and learn logic together.
	(c) A university announcement says new VR training facilities let students practice clinical skills in a safe setting.
Learning reference 0	(a) A promotion invites readers to learn more about an upcoming VR shooting demo.
	(b) A string of coding and technology hashtags that puts machine learning beside VR.
	(c) A film synopsis about a game designer and a marketing trainee trapped inside a VR game.
Tone 3, positive	(a) A player calls a VR horror game great and tells a friend to play it.
	(b) A player calls a VR demo amazing from start to finish and shares a recorded fight.
	(c) A gamer calls a VR title one of the most immersive they have tried.
Tone 2, neutral or mixed	(a) A student asks whether a cabled headset will reach the hallway, since their study room has no space (no evaluation).
	(b) A recap of a console maker's announcement stream lists upcoming games, some in VR, with no evaluative word (no evaluation).
	(c) A player says every VR owner must try one game, but the game needs better filters for eye strain (both).
Tone 1, negative	(a) A player says the motion controllers always bug out, which is frustrating, so they rarely use their VR.
	(b) A writer says they do not like the idea of working and meeting in VR and worries the next generation will stop socializing.
	(c) A buyer says motion sickness will take time to get used to and the headset costs too much for what it plays.

Category	Examples (paraphrased)
Affordance 0, none	(a) A player says they are about to buy the same horror game yet again to play it in VR.
	(b) A recap of a console maker's announcement stream lists upcoming games, some in VR.
Affordance 1, fun or engagement	(a) A player calls a VR horror game great and spooky.
	(b) A gamer calls a VR title one of the most immersive they have tried.
	(c) A player who replaced a broken headset says they are excited to be slicing blocks in a rhythm game again.
Affordance 2, hands-on interaction	(a) A writer wonders whether a VR driving game gives the feel of the road and of acceleration, the control sensations people play for.
	(b) A social VR user says the point of VR for them is that it gets them standing up instead of sitting for hours.
Affordance 3, doing it with others	(a) A player says they want to spend more time in a social VR app with friends.
	(b) A group that cannot meet in person plans to play a tabletop game together in VR.
	(c) A player describes a day of multiplayer VR with children who coached them through a squad game.
Affordance 4, practicing a skill	(a) A VR app promises students practice at interviews, presentations, and public speaking.
	(b) A university announcement says VR training lets students practice clinical skills in a safe setting.
	(c) A player advises a newcomer to practice a VR game in multiplayer first to learn it.
Affordance 5, other benefit	(a) A player tells themselves an all-day rhythm game session counts as exercise.
	(b) A parent says a VR workout app makes them feel powerful again.
	(c) A player says VR has been a big help to them through lockdown.
Constraint 0, none	(a) A player calls a VR horror game great and tells a friend to play it.
	(b) A player calls a VR demo amazing from start to finish.
Constraint 1, physical discomfort	(a) A new headset owner says they are lying down after an hour of a VR tag game because they feel motion sick.
	(b) A player says a cockpit with a lot of roll and pitch is one of the few things that makes them nauseous in VR.
	(c) A player advises newcomers to let themselves get sick, rest, and go back in until the body adjusts.

Category	Examples (paraphrased)
Constraint 2, hardware cost	(a) A player says they play a game without a headset because they cannot afford one.
	(b) A social VR user wants a better headset someday but says they are expensive.
Constraint 3, technical performance	(a) A player says the motion controllers always bug out, so they rarely use their VR.
	(b) A player says their boundary setup resets or puts the floor at the wrong height almost every time they start a VR game.
Constraint 4, other obstacle	(a) A student asks whether a cabled headset will reach the hallway, since their study room has no space.
	(b) A parent says their child skipped a shared VR headset because of germs from everyone wearing it.

Single-coder note

I am the only coder on this study. No second person codes any tweet, so every reliability claim I make is a claim about my own consistency, not agreement between coders. I will recode 15 tweets, 10 percent of the 150, at least 48 hours after the first pass with my first codes hidden, and report percent agreement for each variable.

I am using four safeguards. I wrote every category's inclusion and exclusion rule above before coding the sample. Each category has two or three examples from the archive. I will freeze this codebook before the full pass and log every later change with its date and reason. I code in a fixed order, tweet ID ascending, with the username and like count hidden until the four variables are coded.

With a second coder, I would double-code 30 tweets, 20 percent of the sample, and report Krippendorff's alpha for each variable. I would train the coder on this codebook with 20 practice tweets from outside the sample, then settle each disagreement against the written rule and log it. The finding most at risk is H1, because it rests on tone, the variable that needs the most judgment, and the line between neutral and mixed is where two coders would most often part.

Part 2: Qualitative memo

Before writing the rules, I read 30 tweets drawn at random from the archive, 15 with a learning word and 15 without, and kept them out of the 150 the Sampling Plan will draw. The biggest surprise was how little the learning word means. Only five of the 15 learning-word tweets describe a person learning in VR, such as a player relearning a game on new controllers. The rest used the word for something else: a "learn more" promotion, a film synopsis with a trainee, machine-learning hashtags, and an adult-game advertisement that

passed the archive's sensitive flag. That split is why learning reference is coded by reading, not by the word list.

Most tweets carried no evaluation of VR at all. The praise I did see went to one VR game, a demo or a horror title, rather than to VR itself, which is why tone counts a game played in VR as a VR experience; the few negatives were about setup trouble or about working in VR. Benefits were rare: fun, couch co-op with others, the feel of the controls, practicing on a crane simulator, and one player saying VR helped them through lockdown. Those became affordance categories 1 to 5. Obstacles were motion sickness, including advice on getting used to it, a boundary that kept resetting and a game too demanding for most computers, and a study room with no space for a cabled headset, which became constraint categories 1, 3, and 4. A follow-up search of the archive for price words found players who could not afford a headset, which became category 2.

Two of the 30 were not about VR at all, and digits had been stripped from the text. Each became a decision rule.

Rubric map

The codebook rows total 35 and the memo 15, graded together as one 50-point assignment.

Rubric row	Points	Where the example earns it	Why it earns the full row
Codebook: study overview	3	The study overview paragraph	Three sentences name all three things the row asks for: the question (learning tweets and tone, plus the benefits and obstacles named), the material (English tweets about playing in VR from a named archive and collection route), and the time period (calendar 2021).
Codebook: unit of analysis	8	The unit of analysis paragraph	One coded item is stated exactly, once: one tweet's own text, with what is not part of it named. Every variable is coded on that tweet, and the three looked-up fields describe the same tweet, so nothing is coded on an account or a thread.
Codebook: every variable	5	The four variable tables	Every variable carries all five elements: conceptual definition, operational definition, level of measurement, coded values, and edge-case rules, in that order, with at least four rules each.

Rubric row	Points	Where the example earns it	Why it earns the full row
Codebook: decision rules	7	The seven decision rules	Each one is written as a rule a coder can apply ("code the first one and mark the second a duplicate"), and each covers a hard call met in the material: off-topic matches, a reposted promotion, stripped digits, hashtag strings, advertisements, one clause that fits two variables, and coding blind to the source.
Codebook: examples	5	The examples table	Every category of all four variables has two or three examples, each paraphrased from a real tweet in the archive, none invented, and each clearly fits its category under the rules above it.
Codebook: single-coder note	7	The three single-coder paragraphs	Paragraph one says in plain words that I am the only coder and that every reliability claim is about my own consistency. Paragraph two names four safeguards, above the minimum of three, each from the rubric's list. Paragraph three gives the whole counterfactual: how many tweets, which statistic, how the coder is trained and disagreements settled, and which finding is most at risk and why.
Qualitative memo	15	The three memo paragraphs	About 290 words, inside the 200-to-300-word range, on patterns seen in 30 real tweets rather than the topic in general. Each codebook category traces back to something the memo describes seeing: the learning split to both learning reference values, the unevaluated tweets, the praise for VR games, and the complaints to the three tone values and tone rule 1, the five benefits to affordance categories 1 to 5, the four obstacles and the price search to constraint categories 1 to 4, and the off-topic and stripped-digit tweets to decision rules 1 and 3.

The codebook revises the Definitions Practice entries where the 30 tweets showed a gap: tone now counts a game played in VR as a VR experience, and learning reference gained the rule for relearning a game.